

Q1 - 24 January - Shift 2

The locus of the mid points of the chords of the circle $C_1 : (x - 4)^2 + (y - 5)^2 = 4$ which subtend an angle θ_1 at the centre of the circle C_1 , is a circle of radius r_1 . If $\theta_1 = \frac{\pi}{3}$, $\theta_2 = \frac{2\pi}{3}$ and $r_1^2 = r_2^2 + r_3^2$,

then θ_2 is equal to

(1) $\frac{\pi}{4}$

(2) $\frac{3\pi}{4}$

(3) $\frac{\pi}{6}$

(4) $\frac{\pi}{2}$

Space for your notes:

Q2 - 25 January - Shift 1

The points of intersection of the line $ax + by = 0$, ($a \neq b$) and the circle $x^2 + y^2 - 2x = 0$ are $A(\alpha, 0)$ and $B(1, \beta)$. The image of the circle with AB as a diameter in the line $x + y + 2 = 0$ is :

(1) $x^2 + y^2 + 5x + 5y + 12 = 0$

(2) $x^2 + y^2 + 3x + 5y + 8 = 0$

(3) $x^2 + y^2 + 3x + 3y + 4 = 0$

(4) $x^2 + y^2 - 5x - 5y + 12 = 0$

Space for your notes:

Q3 - 25 January - Shift 2

Points $P(-3,2)$, $Q(9,10)$ and $R(\alpha,4)$ lie on a circle C with PR as its diameter. The tangents to C at the points Q and R intersect at the point S . If S lies on the line $2x - ky = 1$, then k is equal to _____.

Space for your notes:

Q4 - 29 January - Shift 1

Let the tangents at the points $A(4, -11)$ and $B(8, -5)$ on the circle $x^2 + y^2 - 3x + 10y - 15 = 0$, intersect at the point C . Then the radius of the circle, whose centre is C and the line joining A and B is its tangent, is equal to

Space for your notes:

(1) $\frac{3\sqrt{3}}{4}$

(2) $2\sqrt{13}$

(3) $\sqrt{13}$

(4) $\frac{2\sqrt{13}}{3}$

Q5 - 29 January - Shift 2

A circle with centre $(2, 3)$ and radius 4 intersects the line $x + y = 3$ at the points P and Q . If the tangents at P and Q intersect at the point $S(\alpha, \beta)$, then $4\alpha - 7\beta$ is equal to _____.

Space for your notes:

Q6 - 30 January - Shift 1

Let $y = x + 2$, $4y = 3x + 6$ and $3y = 4x + 1$ be three tangent lines to the circle $(x - h)^2 + (y - k)^2 = r^2$. Then $h + k$ is equal to :

(1) 5

(2) $5(1 + \sqrt{2})$

(3) 6

(4) $5\sqrt{2}$

Space for your notes:

Q7 - 30 January - Shift 2

Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C(\sqrt{2}, \sqrt{3})$. Let O

be the origin and OC be perpendicular to both CP

and CQ. If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$,

then $a_1^2 + a_2^2 + b_1^2 + b_2^2$ is equal to _____.

Space for your notes:

Q8 - 31 January - Shift 1

Let a circle C_1 be obtained on rolling the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ upwards 4 units on the tangent T to it at the point $(3, 2)$. Let C_2 be the image of C_1 in T . Let A and B be the centers of circles C_1 and C_2 respectively, and M and N be respectively the feet of perpendiculars drawn from A and B on the x -axis. Then the area of the trapezium $AMNB$ is :

- (1) $2(2 + \sqrt{2})$
- (2) $4(1 + \sqrt{2})$
- (3) $3 + 2\sqrt{2}$
- (4) $2(1 + \sqrt{2})$

Space for your notes:

Q9 - 31 January - Shift 2

The set of all values of a for which the line $x + y = 0$ bisects two distinct chords drawn from a point $P\left(\frac{1+a}{2}, \frac{1-a}{2}\right)$ on the circle

$2x^2 + 2y^2 - (1+a)x - (1-a)y = 0$ is equal to:

- (1) $(8, \infty)$
- (2) $(4, \infty)$
- (3) $(0, 4]$
- (4) $(2, 12]$

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (4) Q2 (1) Q3 (3) Q4 (4)
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Q5 (11) Q6 (1) Q7 (24) Q8 (2)
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Q9 (1)
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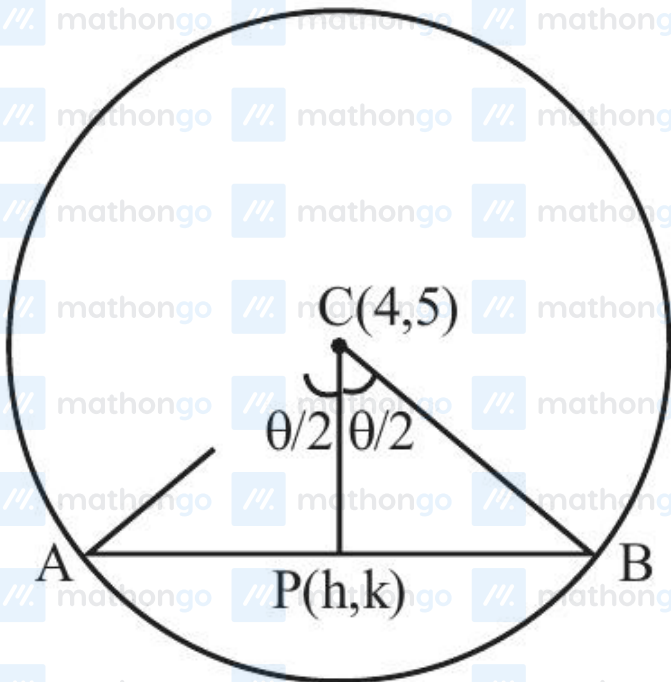
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Q1 (4)							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							

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In $\triangle CPB$



$$\cos \frac{\theta}{2} = \frac{PC}{2} \Rightarrow PC = 2 \cos \frac{\theta}{2}$$

$$\Rightarrow (h-4)^2 + (k-5)^2 = 4 \cos^2 \frac{\theta}{2}$$

$$\text{Now } (x-4)^2 + (y-5)^2 = \left(2 \cos \frac{\theta}{2} \right)^2$$

$$\Rightarrow r_1 = 2 \cos \frac{\pi}{6} = \sqrt{3}$$

$$r_2 = 2 \cos \frac{\theta_2}{2}$$

$$r_3 = 2 \cos \frac{\pi}{3} = 1$$

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$$\Rightarrow r_1^2 = r_2^2 + r_3^2$$

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Questions with Solutions

$$\Rightarrow 3 = 4 \cos \frac{\theta_2}{2} + 1$$

Q2 (1)

Only possibility $\alpha = 0, \beta = 1$

$\therefore$ equation of circle $x^2 + y^2 - x - y = 0$

Image of circle in $x + y + 2 = 0$ is

$$x^2 + y^2 + 5x + 5y + 12 = 0$$

Q3 (3)

$$\Rightarrow \theta_2 = \frac{\pi}{2}$$

$$\Rightarrow \frac{10-2}{9+3} \times \frac{10-4}{9-\alpha} = -1 \Rightarrow \boxed{\alpha = 13}$$

$$y - 10 = -\frac{4}{7}(x - 9)$$

$$m_{OR} \cdot m_{RS} = -1 \Rightarrow m_{RS} = -8$$

$$y - 4 = -8(x - 13)$$

$$x_1 = \frac{25}{2} \quad y_1 = 8$$

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$$25 - 8k = 1$$

$$\Rightarrow 8k = 24$$

Q4 (4)

Equation of tangent at A (4, -11) on circle is

$$\Rightarrow 4x - 11y - 3\left(\frac{x+4}{2}\right) + 10\left(\frac{y-11}{2}\right) - 15 = 0$$

$$\Rightarrow 5x - 12y - 152 = 0 \dots (1)$$

Equation of tangent at B (8, -5) on circle is

$$\Rightarrow 8x - 5y - 3\left(\frac{x+8}{2}\right) + 10\left(\frac{y-5}{2}\right) - 15 = 0$$

$$\Rightarrow 13x - 104 = 0 \Rightarrow x = 8$$

$$\text{put in (1)} \Rightarrow y = \frac{28}{3}$$

$$r = \left| \frac{3 \cdot 8 + \frac{2 \cdot 28}{3} - 34}{\sqrt{13}} \right| = \frac{2\sqrt{13}}{3}$$

Q5 (11)

The given line is polar or $P(2, \beta)$ w.r.t. given circle

$$x^2 + y^2 - 4x - 6y - 3 = 0$$

Chord or contact

$$\alpha x + \beta y - 2(x + \alpha) - 3(y + \beta) - 3 = 0$$

$$\Rightarrow (\alpha - 2)x + (\beta - 3)y - (2\alpha + 3\beta + 3) = 0 \dots (i)$$

$\therefore$ But the equation of chord of contact is given

$$\text{as : } x + y - 3 = 0 \dots (ii)$$

comparing the coefficients

$$\frac{\alpha - 2}{1} = \frac{\beta - 3}{1} = -\left(\frac{2\alpha + 3\beta + 3}{-3}\right)$$

On solving $\alpha = -6$

$$\beta = -5$$

Now $4\alpha - 7\beta = 11$

Q6 (1)

$$L_1 : y = x + 2, L_2 : 4y = 3x + 6, L_3 : 3y = 4x + 1$$

Bisector of lines L_2 & L_3

$$\frac{4x - 3y + 1}{5} = \pm \left(\frac{3x - 4y + 6}{5} \right)$$

$$(+)\ 4x - 3y + 1 = 3x - 4y + 6$$

$$x + y = 5$$

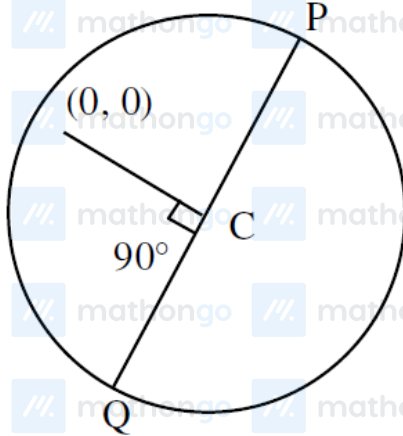
Centre lies on Bisector of $4x - 3y + 1 = 0$ &

$$(0)\ 3x - 4y + 6 = 0$$

$$\Rightarrow h + k = 5$$

Q7 (24)

$$\frac{1}{2} \times PC \times \sqrt{5} = \frac{\sqrt{35}}{2}; PC = \sqrt{7}$$



$$a_1^2 + b_1^2 + a_2^2 + b_2^2 = OP^2 + OQ^2$$

$$= 2(5 + 7) = 24$$

Q8 (2)

$$C = (2, 3), r = \sqrt{2}$$

$$\text{Centre of } G = A = 2 + 4 \frac{1}{\sqrt{2}},$$

$$3 + \frac{4}{\sqrt{2}} = (2 + 2\sqrt{2}, 3 + 2\sqrt{2})$$

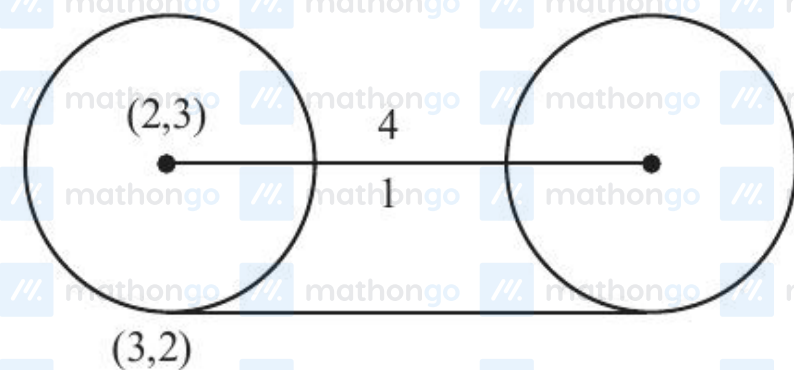
$$A(2 + 2\sqrt{2}, 3 + 2\sqrt{2})$$

$$B(4 + 2\sqrt{2}, 1 + 2\sqrt{2})$$

$$\frac{x - (2 + 2\sqrt{2})}{1} = \frac{y - (3 + 2\sqrt{2})}{-1} = 2$$

$\therefore$ area of trapezium:

$$\frac{1}{2}(4 + 4\sqrt{2})2 = 4(1 + \sqrt{2})$$



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Circle

Questions with Solutions

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Q9 (1)

The diagram shows a circle with a horizontal line passing through its center. Two vertical lines are drawn perpendicular to the horizontal line, intersecting it at points M and N. The points M and N are marked on the horizontal line. The diagram is overlaid on a grid of 'mathongo' watermarks.

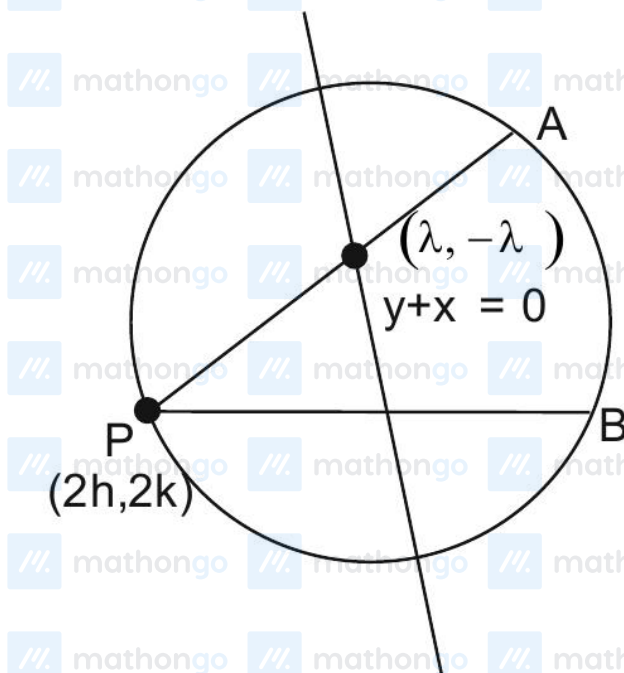
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$$x^2 + y^2 - \frac{(1+a)x}{2} - \frac{(1-a)y}{2} = 0$$

$$\text{Centre } \left(\frac{1+a}{4}, \frac{1-a}{4} \right) \Rightarrow (h, k)$$

$$P \left(\frac{1+a}{2}, \frac{1-a}{2} \right) \Rightarrow (2h, 2k)$$



$$\begin{aligned} \text{Equation of chord} &\Rightarrow T = S_1 \\ &\Rightarrow (x-y)\lambda - \frac{2h(x+\lambda)}{2} - \frac{(2k)(y-\lambda)}{2} \end{aligned}$$

$$= 2\lambda^2 - 2h\lambda -$$

Now, $\lambda - 2h$ satisfies the chord

$$\therefore (2h - 2k)\lambda - h(x + \lambda) - k(y - \lambda)$$

$$\Rightarrow 2\lambda^2 + 4k\lambda - 4h\lambda + h\lambda - k\lambda + hx + ky = 0$$

$$\Rightarrow 2\lambda^2 + \lambda(3k - 3h) + ky + hx = 0$$

$$\Rightarrow D > 0$$

$$\Rightarrow 9(k-h)^2 - 8(ky + hx) > 0$$

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$$\Rightarrow 9(k-h)^2 - 8(2k^2 + 2h^2) > 0$$

Questions with Solutions

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$$\Rightarrow 7k^2 + 7h^2 + 18kh < 0$$

$$\Rightarrow 7\left(\frac{1-a}{4}\right)^2 + 7\left(\frac{1+a}{4}\right)^2 + 18\left(\frac{1-a^2}{16}\right) < 0$$

$$\Rightarrow 7\left[\frac{2(1+a^2)}{16}\right] + \frac{18(1-a^2)}{16} < 0, \quad a^2 = t$$

$$\Rightarrow \frac{7}{8}(1+t) + \frac{18(1-t)}{16} < 0$$

$$\Rightarrow \frac{14 + 14t + 18 - 18t}{16} < 0$$

$$\Rightarrow 4t > 32$$

$$t > 8 \quad a^2 > 8$$

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